

Serhat Belen

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building backend systems, AI-powered workflows, and data-intensive web applications in production. My core focus areas are Node.js / NestJS / Express, event-driven architectures (RabbitMQ), real-time processing pipelines, and LLM API integrations. On the frontend, I build SEO-friendly, SSR / SSG-enabled interfaces with React and Next.js, with close attention to state management, UI/UX, and animation details. I have hands-on experience in healthcare SaaS: I built agentic architectures that automate insurance workflows, diagnosed and resolved Node.js memory leak issues in production, and optimized large-scale MongoDB aggregation pipelines. In fast-moving startup environments, I take end-to-end ownership of architectural decisions and can quickly learn and adapt to new languages and frameworks whenever needed.

TECHNICAL SKILLS

Backend:	Node.js, NestJS, Express, REST APIs, Event-Driven Architecture
Frontend:	React, Next.js (SSR / SSG), React Native, Vue.js, TypeScript, Tailwind CSS, Shadcn UI, Redux, Zustand, Jotai, UI/UX, Animations (Framer Motion, Reanimated)
AI / LLM:	LLM API Integration, Agentic Workflow Design, Prompt Engineering, RAG, Embedding-Based Search
Database:	MongoDB & Mongoose (Aggregation Pipelines, Indexing, Schema Design), PostgreSQL, Supabase, TypeORM, Redis, NoSQL / SQL Optimization
Infrastructure:	Docker, Dockerized Monorepo Deployments, AWS (EC2, S3, Lightsail), CI/CD, Nginx, Ubuntu Server
Messaging:	RabbitMQ, Queue Architecture, Async Processing Pipelines
Engineering:	System Design, Plugin Architecture, Memory Profiling, Performance Optimization, Unit Testing (Jest)
Open to:	New languages and runtimes; I quickly learn and adapt to whatever a project requires

WORK EXPERIENCE

Full Stack Developer

Jun 2025 - Jun 2026

Onlay AI

React, NestJS, MongoDB, PostgreSQL, RabbitMQ, Redis, AWS, LLM APIs

- Designed and built the architecture of an agentic automation platform that automates dental insurance workflows end to end, unifying real-time eligibility verification, claim preparation and submission, denial management, EOB parsing, payment posting, and AR follow-up into a single integrated system.
- Designed document-understanding pipelines that allow the system to independently read EOBs and insurance coverage documents, extract structured benefit data, and trigger follow-up actions without human intervention when needed.
- Migrated to an event-driven processing architecture using RabbitMQ, splitting tasks such as eligibility verification, claim submission, and denial detection into independently scalable consumers, cutting processing latency by roughly half.

- Built real-time eligibility verification pipelines that automate benefit parsing and coverage detection, significantly reducing the front-office team's workload.
- Built an automated claims submission engine combining supplemental document extraction, claim packaging, submission to the insurance carrier, and retry / denial detection logic.
- Developed multi-purpose operational dashboards that let clinics track treatment processes, earnings, denied procedures, payments, ERA processes, and auto-posting from a single place.
- Built the frontend of these dashboards with React and TypeScript, using Zustand for state management, Shadcn UI / Radix primitives for accessible components, and Tailwind CSS for a consistent design language.
- Root-caused and resolved critical production memory leak issues through Node.js heap profiling and service-level debugging, stabilizing services under high concurrency.
- Owned backend architectural decisions for fully asynchronous insurance processing pipelines, defining service boundaries and data flow contracts.

Full Stack Developer

Apr 2021 - May 2025

Avena - Retrace

React, Node.js, MongoDB, Redis, RabbitMQ, Puppeteer, Docker

- Developed high-throughput EDI services for distributed clinical systems, processing millions of health records per year.
- Optimized complex MongoDB aggregation pipelines by restructuring stage order, adding compound indexes, and denormalizing data, noticeably reducing reporting query times.
- Improved application performance with Redis caching strategies, reducing dashboard response times by roughly 40%.
- Built a Puppeteer-based automation for insurance eligibility verification, reducing manual workload and improving data accuracy.
- Converted detailed, rich interface designs from designers into reusable, component-based React code, making heavy use of custom hooks to separate business logic from the UI.
- Developed similar multi-purpose operational dashboards showing clinics' treatment processes, revenue, denied procedures, and payment tracking, using Redux for state management.
- Deployed and maintained Dockerized monorepo architectures on AWS EC2 and Lightsail, including Nginx configuration and Ubuntu server administration.
- Built processes for saving, reading, and updating attachments in different formats (DICOM, PDF, CSV) with organized folder structures on AWS S3.

Frontend Developer

Sep 2020 - Apr 2021

Bynogame

jQuery, JavaScript, Laravel APIs, SCSS

- Built performance-focused interfaces for a high-traffic e-commerce platform serving hundreds of thousands of users monthly.
 - Converted Figma designs into responsive, production-ready UI components used throughout the application.
 - Developed real-time product feed pages with instantly updating price and stock information, efficiently managing DOM updates with jQuery.
 - Contributed to frontend performance and SEO optimizations, achieving measurable improvements in page load times.
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PROJECTS

Ciftopia - Couples App

React Native / Expo · NestJS · Socket.io · OpenAI · Gemini TTS · React Three Fiber · RevenueCat · AWS S3

A full-stack mobile app I built for couples. Features a real-time quiz engine running over Socket.io, AI-generated stories narrated with OpenAI and Gemini TTS, and a gamification system that evolves over time through shared content, along with a 3D space explorer built with React Three Fiber. I manage subscriptions and in-app purchases through RevenueCat, use AWS S3 for media storage, and Expo / FCM for notifications.

Dental B2B Marketplace

Next.js · Zustand · PostgreSQL

A B2B dental supplies platform connecting dental clinics with approved suppliers across 8+ categories and 50,000+ products. Built the frontend with Next.js focused on SSR / SEO, and the backend on a REST API; applied compound indexes and query optimizations in PostgreSQL to comfortably handle the high product volume.

Zikirmatik - Dhikr App

React Native · NestJS · Turborepo · OpenAI GPT-4o-mini · Text Embeddings

A full-stack mobile app built on a carefully curated dataset of roughly 500 dhikr entries. Using text embedding-based semantic search, it recommends the most contextually relevant dhikr with GPT-4o-mini based on a user's natural-language query. Used Turborepo for monorepo management and NestJS on the API side; integrated OpenAI for embedding generation and AI-powered recommendations.

Time Heroes - Idle Pet-Collecting RPG

React · TypeScript · Jotai

Developed the in-game interface and client-side game logic for an idle RPG. Built interactive game screens with React and TypeScript, and handled state management across components with Jotai.

EDUCATION & LANGUAGES

Cukurova University, Computer Engineering, B.Sc. (2013 - 2019)

Turkish: Native | English: Business English